

1 Making Decisions under Uncertainty

2 Convergent Prior

Definition 2.1 (probabilistic model). We define a *probabilistic model* as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of *acceptable* probability measures on $\mathcal{S}$.

Definition 2.2 (ϵ -convergence). Let S and E be random variables in a probabilistic model $\mathcal{M}$. We say $S|E$ (S given E) ϵ -converges by a distribution P in $\mathcal{M}$, if for every distribution Q in $\mathcal{M}$ we have:

$$\mathbb{E}_Q[\mathbb{D}(P_{S|E} || Q_{S|E})] = \sum_{s \in \mathcal{S}} \sum_{e \in \mathcal{E}} \ln \frac{Q(s | e)}{P(s | e)} Q(s, e) < \epsilon . \quad (1)$$

Definition 2.3 (ϵ -convergent distribution). Let S be a random variable, and $\mathbf{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. We say P is an ϵ -convergent distribution for $S|\mathbf{E}$ in $\mathcal{M}$, if for all $E \in \mathbf{E}$, $S|E$ ϵ -converges in $\mathcal{M}$ by P .

Theorem 1. Consider two probabilistic models $\mathcal{M}$ and $\mathcal{M}^*$. Let $\mathbf{E}$ be a countable set of random variables in $\mathcal{M}$, and E^* be a single random variable in $\mathcal{M}^*$. Assume there exist a random variable I in $\mathcal{M}^*$, such that for every probability measure Q in $\mathcal{M}$, there exist a probability measure Q^* in $\mathcal{M}^*$, such that for every $E_i \in \mathbf{E}$ we have:

$$Q(S, E_i) = Q^*(S^*, E^* | I = i) , \quad (2)$$

while I and E^* are conditionally independent:

$$Q^*(S^* | E^*) = Q^*(S^* | E^*, I) . \quad (3)$$

Then $S|\mathbf{E}$ ϵ -converges by P in $\mathcal{M}$, if $S^*|E^*$ ϵ -converges by P^* in $\mathcal{M}^*$.

Proof. Assume, by contradiction, that $S|\mathbf{E}$ does not ϵ -converge by P in $\mathcal{M}$. Therefore, there must exist Q and $E_i \in \mathbf{E}$ in $\mathcal{M}$ such that:

$$\mathbb{E}_Q[\mathbb{D}(P_{S|E_i} || Q_{S|E_i})] \geq \epsilon .$$

Equations 2 and 3 hold for every Q and E_i , and we can rewrite the left hand

side of this inequality using them:

$$\begin{aligned}
\mathbb{E}_Q[\mathbb{D}(P_{S|E_i} || Q_{S|E_i})] &= \sum_{s \in \mathcal{S}} \sum_{e_i \in \mathcal{E}_i} \ln \frac{Q(s | e_i)}{P(s | e_i)} Q(s, e_i) \\
&= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*, I = i)}{P^*(s^* | e^*, I = i)} Q^*(s^*, e^* | I = i) \\
&= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^* | I = i) ,
\end{aligned}$$

so we must have:

$$\sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^* | I = i) \geq \epsilon .$$

Since $Q^*(i)$ is a distribution and is non-negative for every i , we can multiply both sides of this inequality to $Q^*(i)$ and sum over all values of i .

$$\begin{aligned}
\sum_i \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^*, I = i) &\geq \sum_i \epsilon Q^*(i) \\
\sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} \sum_i Q^*(s^*, e^*, I = i) &\geq \epsilon \sum_i Q^*(i) \\
\sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* | e^*)}{P^*(s^* | e^*)} Q^*(s^*, e^*) &\geq \epsilon ,
\end{aligned}$$

which means $S^*|E^*$ does not ϵ -converge by P^* in $\mathcal{M}^*$. $\square$

Definition 2.4 (conditional). A *conditional* is a set of conditional probabilities of an event A such that the conditions form a partition of the sample space, i.e they are disjoint events whose their probabilities sum to one.

$$P(A | XE_i) | E_i \cap E_j = \emptyset$$

Note that according to this definition, an event conditioned on a vector of random variables is always a conditional.

Kullback-Leibler directed divergence

$$< \sum_{s \in \mathcal{S}} \sum_{e \in \mathcal{E}} \ln \frac{Q(s | e)}{P(s | e)} Q(s, e) < \epsilon \tag{4}$$

lkllklk said Nobody [1]

References

- [1] Nobody Jr. My article, 2006.